

**Major Project Stage-1 First Review Presentation
on**

**Predictive Thermal Battery Management System
for Li-ion Battery Packs**

Presented by

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Guide

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- Recap of the zeroth review
- Problem statement and objectives
- Literature survey
- Proposed system, hardware and circuits
- Firmware and the predictive algorithm
- IoT dashboard, cost and work plan

Recap of the Zeroth Review

- The idea was approved: a BMS that acts on how fast a cell heats, not only on how hot it is
- The synopsis and the block diagram were accepted
- My guide signed off, so Phase 1 of the plan is complete
- Since then I have fixed the circuit values and priced every part
- This review covers the detailed design and the firmware plan

Problem Statement

- A normal small BMS cuts power only when a fixed temperature is crossed, often 60 °C
- A cell that is climbing fast at 35 °C is already faulty, but it passes that test
- Once a lithium cell starts heating itself, the heat drives more reaction, which makes more heat
- A sensor on the outside of a cell always lags what is happening inside it
- So a fixed limit reacts late, and it gives no warning at all beforehand

Objectives

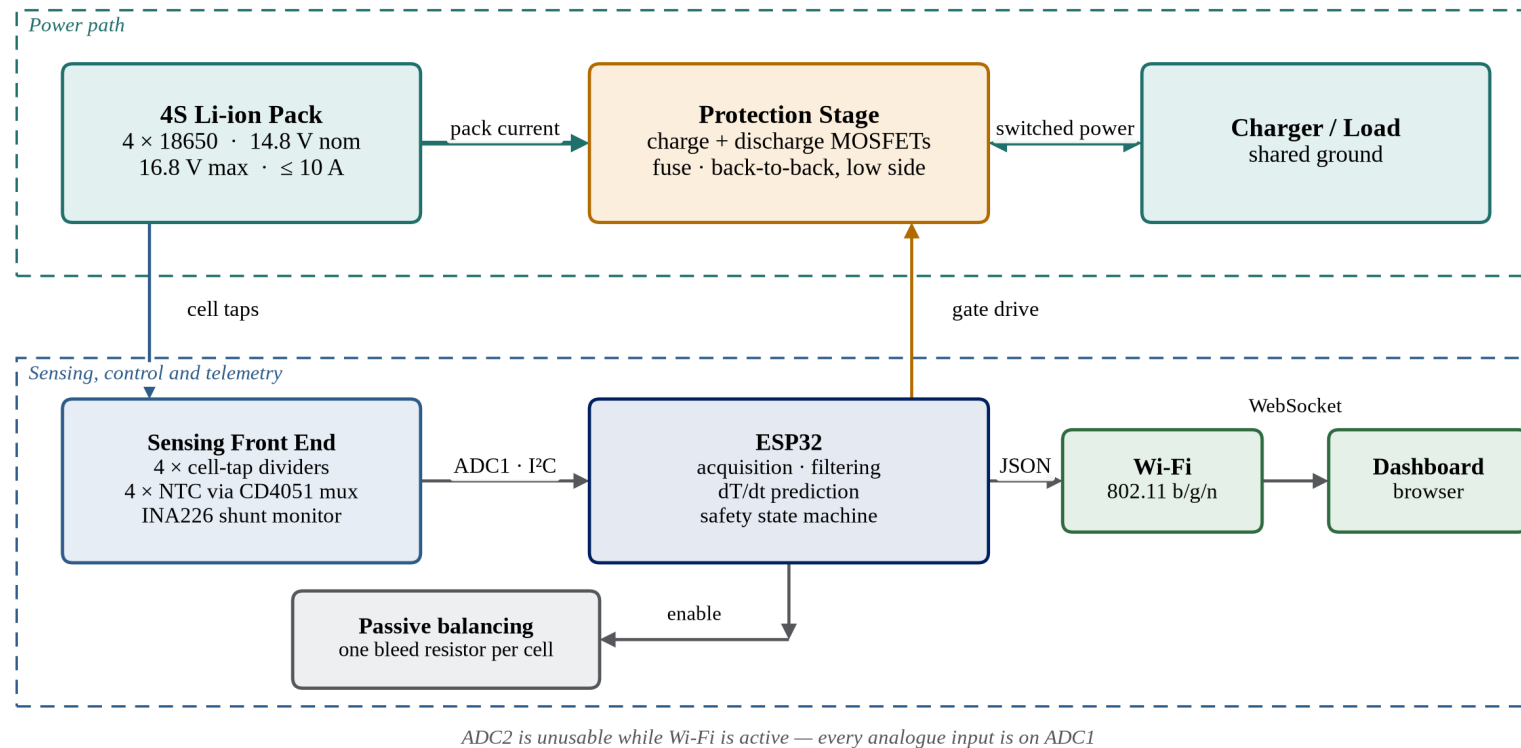
- Build a working 4S battery management system for 18650 cells
- Measure four cell voltages, the pack current and four cell temperatures
- Trip on the rate at which temperature rises, not only on its value
- Switch charging and discharging separately, so only the unsafe direction stops
- Send every reading over Wi-Fi to a page that opens on a phone or a laptop

Literature Survey

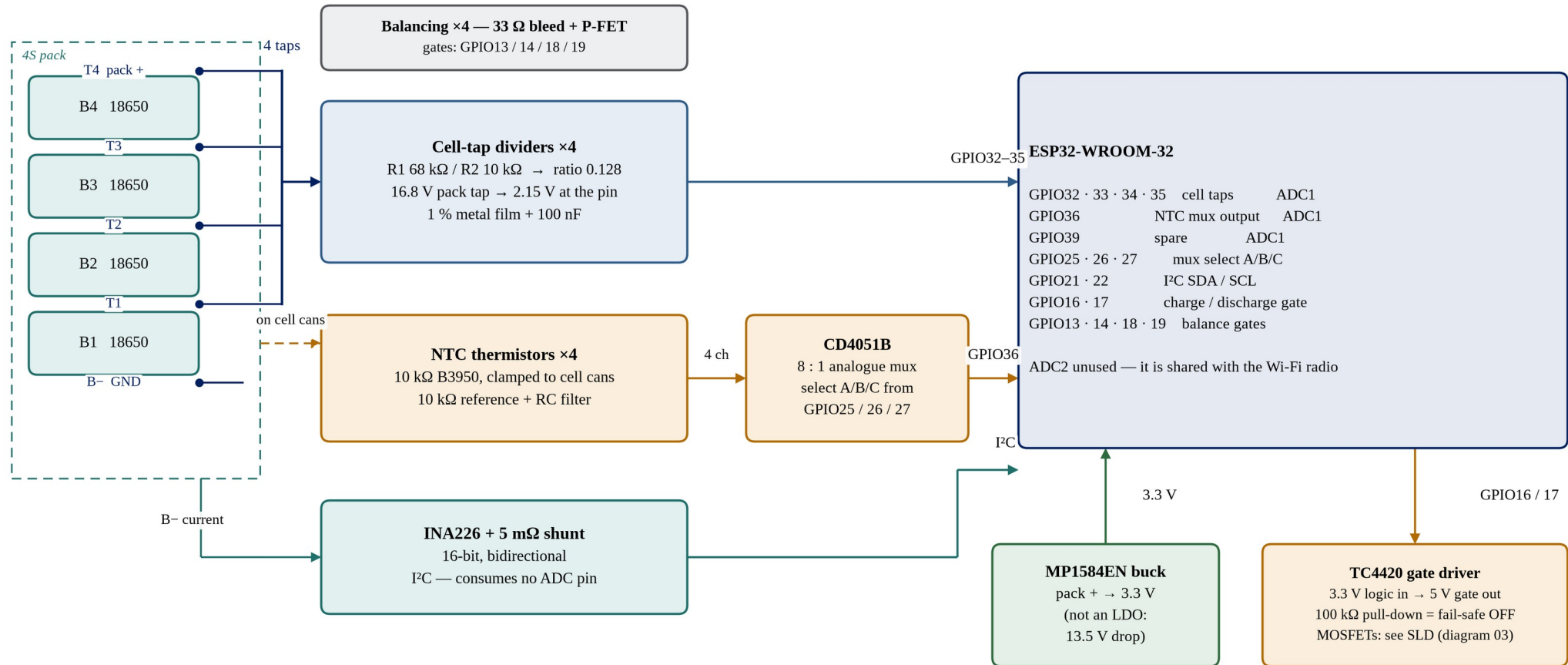
Ref	Author and year	What they did	What it does not do	What I do differently
[1]	Feng et al., 2018	Laboratory study of how runaway starts and spreads	Explains the mechanism; it is not a live detector	Turns that mechanism into a trip that runs on the board
[2]	Zhang et al., 2023	Review of ways to predict runaway early	Most methods need gas or pressure sensors, or a server	One threshold test on the ESP32, with no extra sensors
[3]	Chen et al., 2024	Trips when the temperature rise reaches 1 °C per second	Needs training data and a fitted thermal model	Uses the same 1 °C/s figure, computed directly in firmware
[4]	Habib et al., 2023	Review of BMS functions and the problems still open	Lists thermal runaway as unsolved; builds nothing	Builds and bench-tests one specific answer to it

Proposed System

- The pack, the sensing, the controller and the protection sit in one loop
- The same readings are used to protect the pack and to feed the dashboard



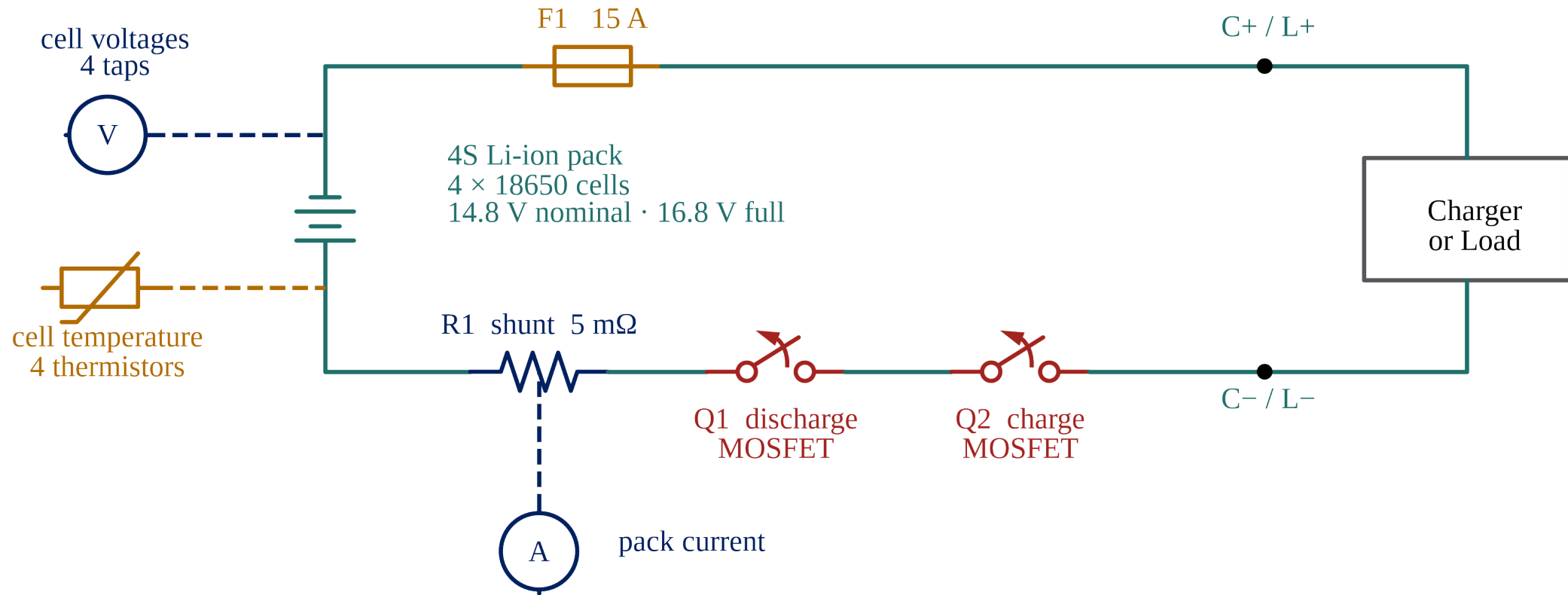
Hardware Architecture



4 cell taps + 4 thermistors = 8 analogue channels, but a DevKit exposes only 6 ADC1 pins — so the thermistors are multiplexed

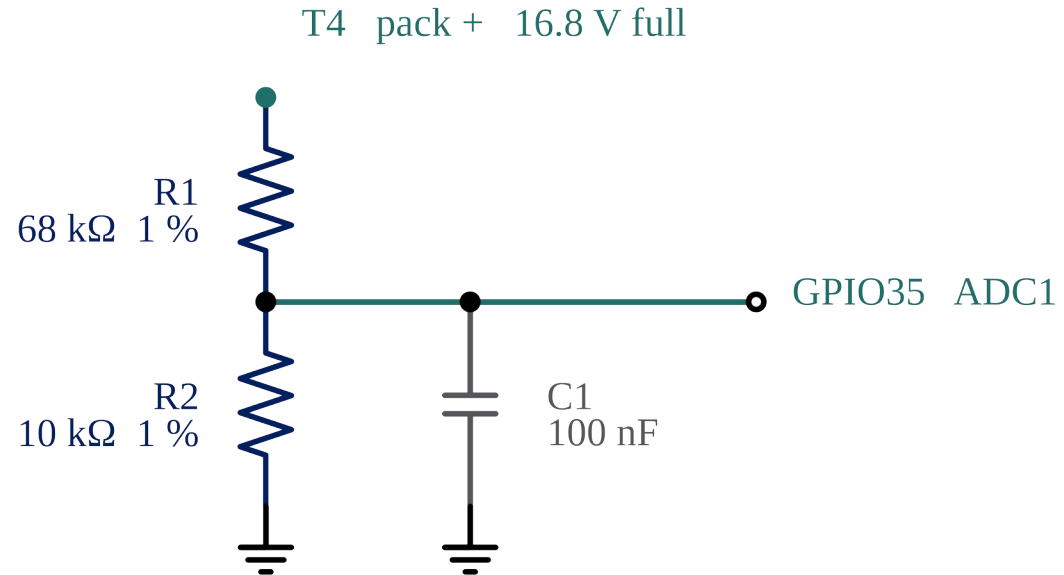
Power Path — Single Line Diagram

- The 15 A fuse is the backstop if the electronics fail completely
- Both switches sit in the negative return, so both gates drive against ground

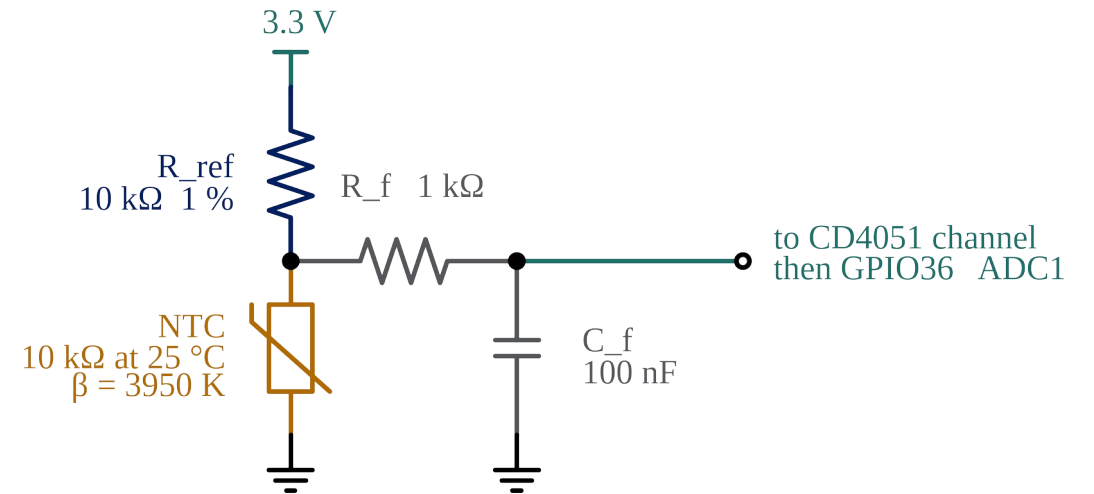


Sensing Circuits

Cell voltage



Cell temperature

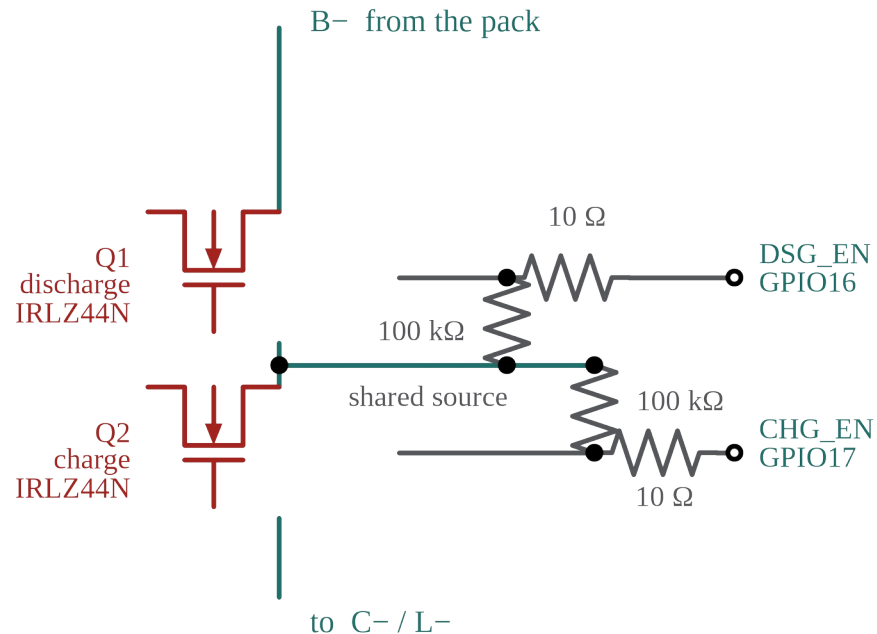


- Divider ratio 0.128 puts a full 16.8 V pack at 2.15 V — about 4.7 mV per count
- Beta equation: $1/T = 1/T_0 + (1/\beta) \cdot \ln(R/R_0)$, with $T_0 = 298.15$ K, $R_0 = 10$ k Ω , $\beta = 3950$ K

Protection and Balancing

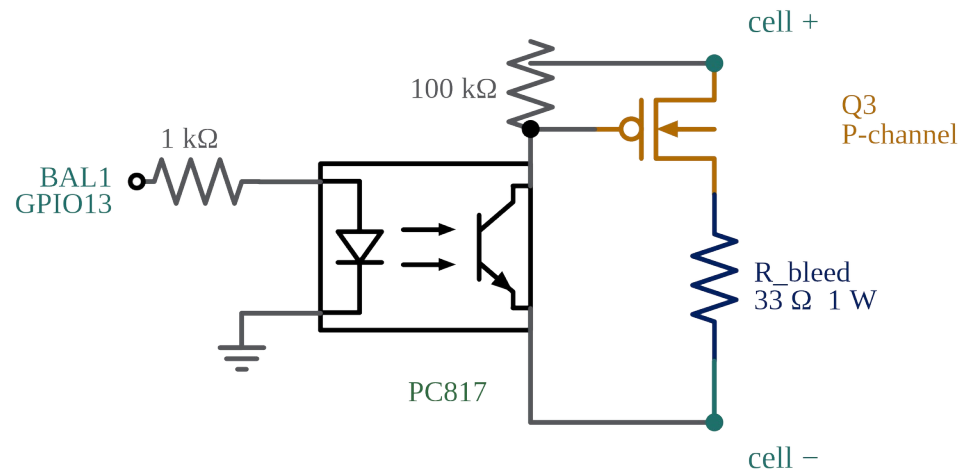
- IRLZ44N: logic level, 55 V, about 22 mΩ on-resistance at a 5 V gate
- The two body diodes face opposite ways, so each direction blocks separately

Cut-off pair — low side, back to back



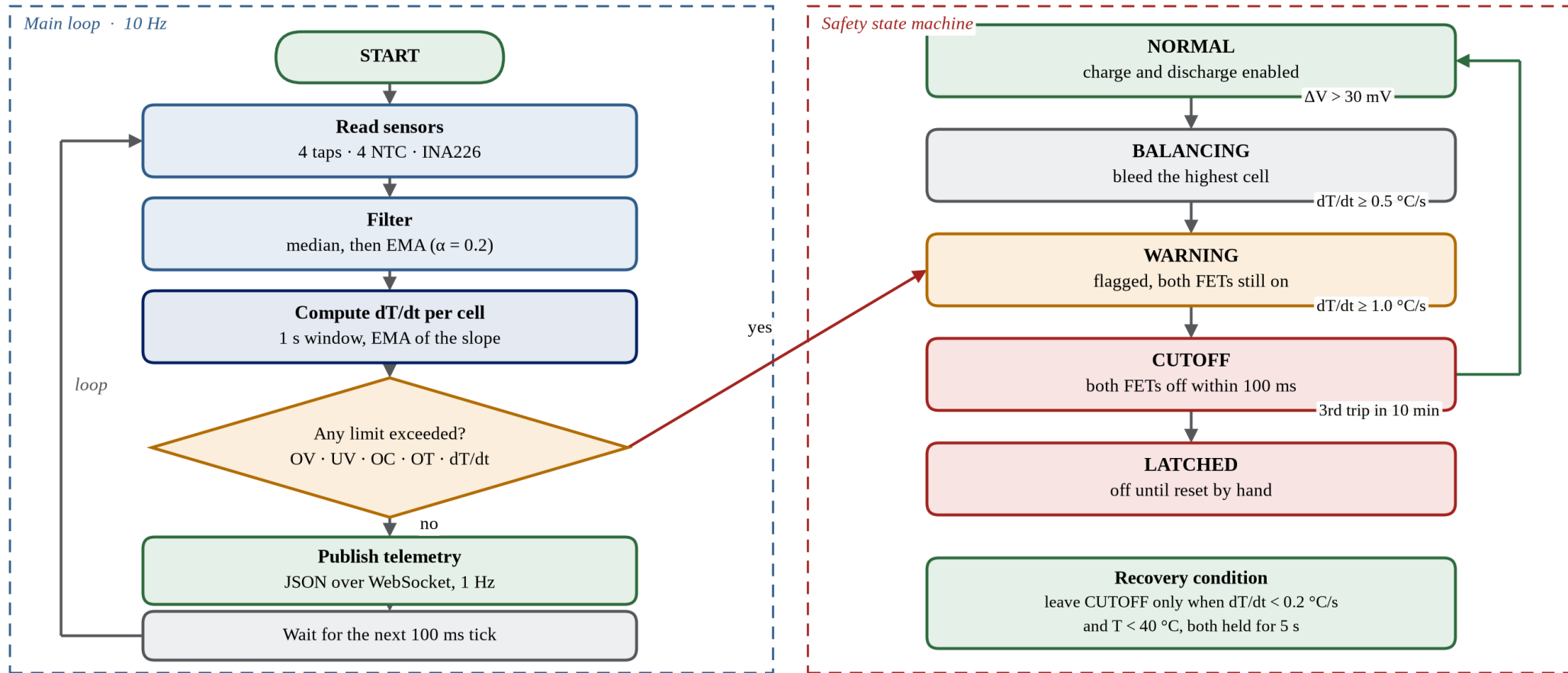
The 100 kΩ resistors hold both gates at the shared source, so the pair sits OFF whenever the ESP32 is not driving it.

Passive balancing — one cell



The optocoupler keeps the drive isolated, so one GPIO referenced to ground can switch a cell that sits several volts above ground.
Bleed current at 4.0 V is $4.0 / 33 = 121$ mA, about 0.48 W.

Firmware Architecture



The Predictive Algorithm

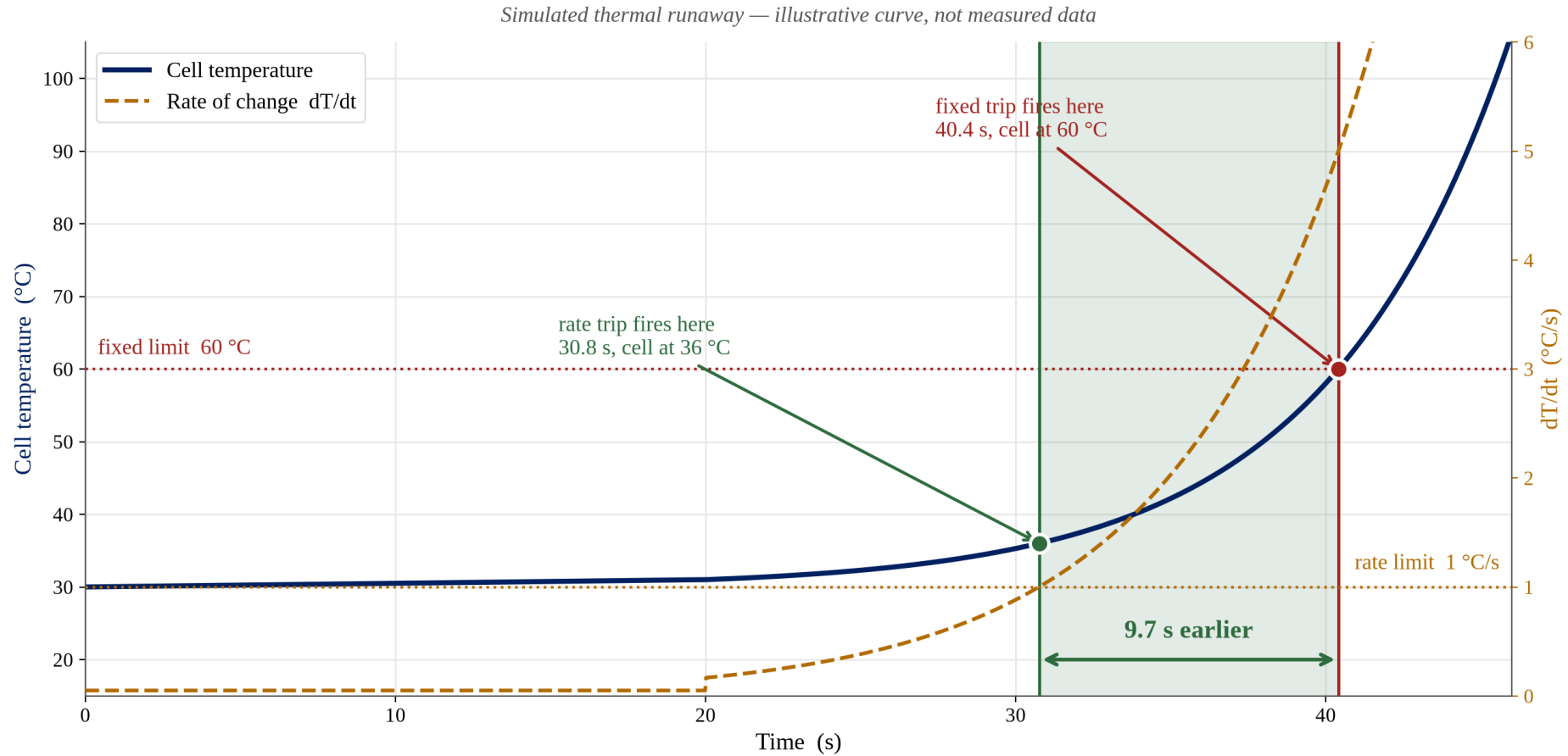
- Temperature is sampled every 100 ms and smoothed before the slope is taken
- The slope is an exponential moving average over a one-second window

```
T_filt = ema(T_raw, alpha = 0.2)
slope  = (T_filt - T_filt_1s_ago) / 1.0
dTdt   = ema(slope, alpha = 0.3)

if dTdt >= 1.0 or T_filt >= 60.0:
    open_both_mosfets()
    state = CUTOFF
elif dTdt >= 0.5:
    state = WARNING
# leave CUTOFF only when dTdt < 0.2 and T_filt < 40 for 5 s
```

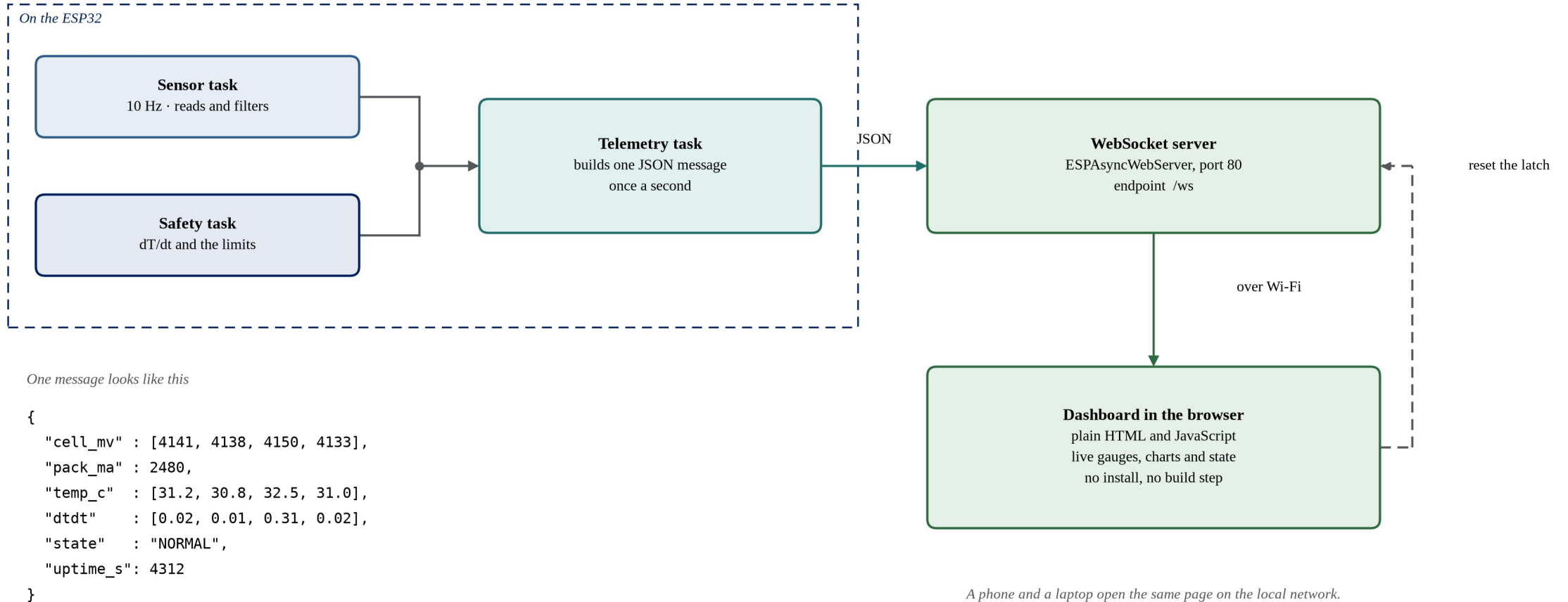
- Smoothing comes first: a slope taken from raw counts is mostly noise
- A fixed calibration error cancels in a slope, so the trip tolerates ADC drift

Predictive Compared With a Fixed Limit



Simulated for illustration. Real measurements will replace this after bench testing.

IoT Dashboard



Bill of Materials

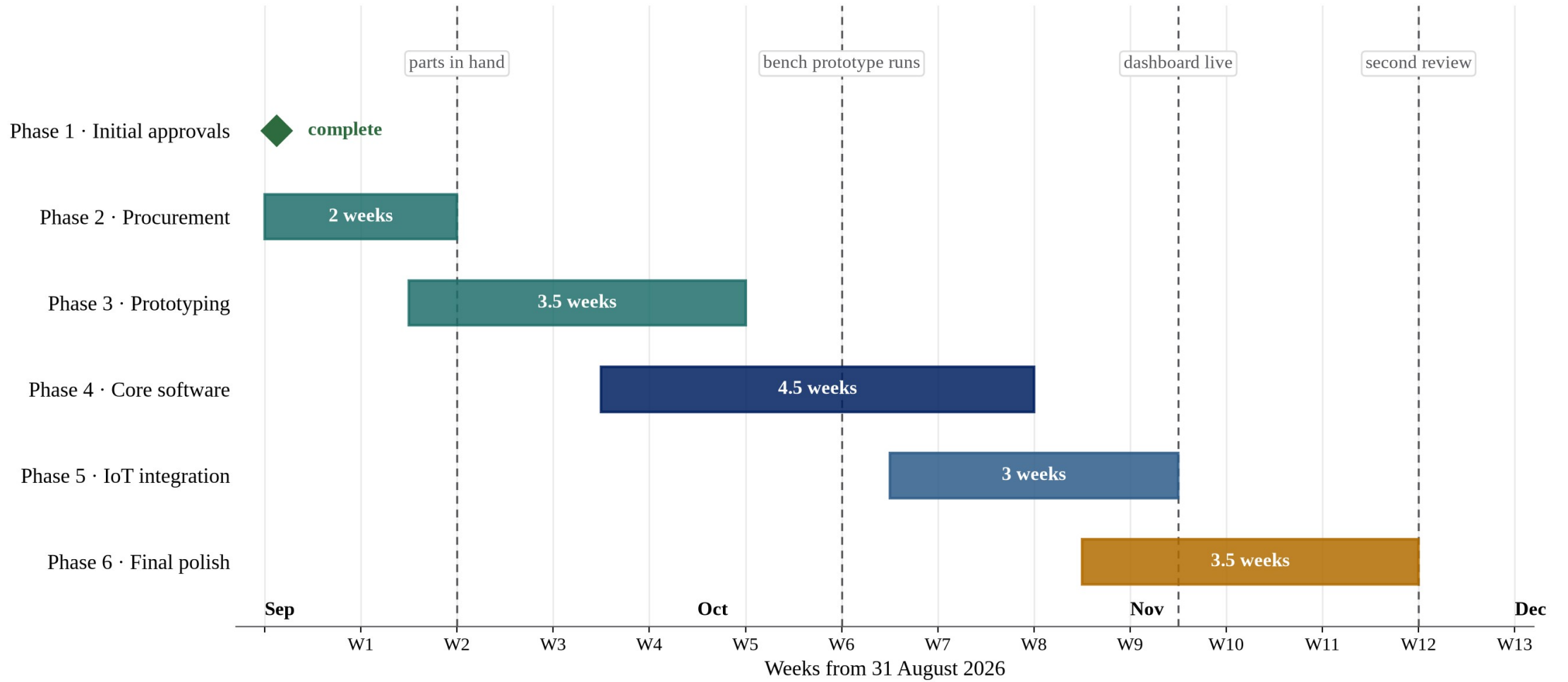
Approximate Indian retail prices, August 2026

Item	Qty	Unit (₹)	Amount (₹)
ESP32-WROOM-32 DevKit V1	1	450	450
18650 cells, 2600 mAh	4	250	1000
4S holder and nickel strip	1 set	270	270
NTC thermistor 10 kΩ B3950	4	7	28
INA226 module with shunt	1	250	250
IRLZ44N cut-off MOSFETs	2	45	90
Balancing parts per cell (P-FET, opto, bleed)	4 sets	45	180
CD4051B multiplexer and TC4420 driver	1 set	145	145
MP1584EN buck module	1	90	90
Passives, fuse, board and wiring	1 set	550	550
Total			3053

Work Completed

Phase	Status	What is done
1 · Initial approvals	Complete	Synopsis, block diagram and guide approval
2 · Procurement	In progress	Parts list finalised and priced; ordering next
3 · Prototyping	Not started	Waiting on parts
4 · Core software	Not started	Algorithm and state machine designed on paper
5 · IoT integration	Not started	Message format decided
6 · Final polish	Not started	—

Work Plan



Expected Outcomes

- A working 4S BMS that cuts off on temperature rate, demonstrated on the bench
- Measured proof that the rate trip fires earlier than a fixed 60 °C limit
- A live dashboard showing voltage, current, temperature and rate for every cell
- A design that costs about ₹3,000 and uses only parts available in India

References

- [1] X. Feng, M. Ouyang, X. Liu, L. Lu, Y. Xia and X. He, “Thermal runaway mechanism of lithium ion battery for electric vehicles: A review,” *Energy Storage Materials*, vol. 10, pp. 246–267, 2018.
- [2] X. Zhang, S. Chen, J. Zhu et al., “A critical review of thermal runaway prediction and early-warning methods for lithium-ion batteries,” *Energy Material Advances*, vol. 4, art. 0008, 2023.
- [3] Q. Chen, Y. He, N. Fang and G. Yu, “A combined data-driven and model-based algorithm for accurate battery thermal runaway warning,” *Sensors*, vol. 24, no. 15, art. 4964, 2024.
- [4] A. K. M. A. Habib, M. K. Hasan, G. F. Issa, D. Singh, S. Islam and T. M. Ghazal, “Lithium-ion battery management system for electric vehicles: Constraints, challenges and recommendations,” *Batteries*, vol. 9, no. 3, art. 152, 2023.
- [5] IS 16046 (Part 2) : 2018 / IEC 62133-2 : 2017, Secondary cells and batteries containing alkaline or other non-acid electrolytes — Part 2: Lithium systems.

Thank You

Questions are welcome